Comprehensive Smart Precision Irrigation Advisory - Wheat Crop

1. Crop Duration

Wheat crop duration ranges from 110–130 days depending on variety and agro■climatic conditions.

2. Growth Stages with Moisture Requirement

Stage	Days After Sowing	Soil Moisture (%)	Remarks
Germination	0–7	70–80	Ensure proper emergence
CRI	20–25	70–80	Most critical stage
Tillering	25–45	65–75	Vegetative growth
Jointing	45–60	65–75	Stem elongation
Booting	60–75	65–70	Spike formation
Flowering	75–90	70–80	High water demand
Milk Stage	90–110	60–70	Grain filling
Maturity	110–130	50–60	Reduce irrigation

3. Irrigation Schedule

Irrigation No.	Stage	Days After Sowing	Water Depth (mm)	Approx Duration (Drip)
1	CRI	20–25	60–70	2–3 hrs
2	Tillering	40–45	60–70	2–3 hrs
3	Jointing	60–65	60–70	2–3 hrs
4	Booting/Flowering	80–85	60–70	3–4 hrs
5	Milk Stage	100–105	60–70	2–3 hrs
6 (Optional)	Dough Stage	115–120	50–60	2 hrs

4. Fertilizer Application Based on Growth Stage

Stage	Fertilizer	Dose (kg/ha)	Purpose
Sowing (Basal)	DAP / NPK	N:60, P:60, K:40	Root establishment
CRI Stage	Urea	30 kg N	Tillering improvement
Jointing Stage	Urea	30 kg N	Stem & spike growth
Before Sowing	Zinc Sulphate	25	Micronutrient correction

5. Total Seasonal Water Requirement

Total water requirement for wheat crop is approximately 450–650 mm during the complete crop cycle. Irrigation should be scheduled using soil moisture sensors for precision irrigation systems.